

Analog – Problem 1.1.56

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Problem Statement

Problem 1.1.56

Consider the triangular wave generator shown in Fig. 1.1.56.1. Assume ideal op-amps with $\pm 2\text{ V}$ supply. The input is a $+5\text{ V}$, 50 Hz square wave with 50% duty cycle.

The condition that results in a triangular wave of 5 V peak-to-peak and 50 Hz at the output is:

- ① $RC = 1$
- ② $R/C = 1$
- ③ $R/C = 5$
- ④ $C/R = 5$

Circuit Diagram

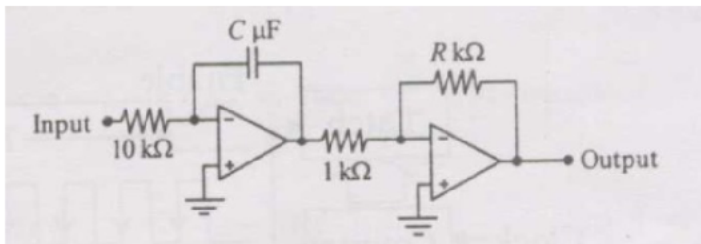


Fig : Circuit

Step 1: Fourier Series of the Square Wave

The input is a bipolar square wave with amplitude $V_p = 5\text{ V}$, fundamental $\omega_0 = 2\pi f_0 = 100\pi\text{ rad/s}$.

The Fourier series coefficients b_n are computed as:

$$b_n = \frac{2}{T} \int_0^T V_{\text{in}}(t) \sin(n\omega_0 t) dt$$

Since the square wave is odd and has half-wave symmetry, only odd n contribute, giving:

$$b_n = \begin{cases} \frac{4V_p}{n\pi} & n = 1, 3, 5, \dots \\ 0 & n \text{ even} \end{cases}$$

The complete Fourier series is:

$$V_{\text{in}}(t) = \frac{4V_p}{\pi} \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{1}{n} \sin(n\omega_0 t) = \frac{4V_p}{\pi} \left[\sin \omega_0 t + \frac{1}{3} \sin 3\omega_0 t + \frac{1}{5} \sin 5\omega_0 t + \dots \right]$$

Step 2: Laplace Transfer Function – Stage 1

The inverting integrator has input impedance $Z_{in}(s) = R_1$ and feedback impedance $Z_f(s) = 1/(sC)$.

For an inverting op-amp configuration:

$$H_1(s) = -\frac{Z_f(s)}{Z_{in}(s)} = -\frac{1/(sC)}{R_1} = -\frac{1}{sR_1C}$$

Evaluating on the imaginary axis $s = jn\omega_0$ for the n -th harmonic:

$$H_1(jn\omega_0) = -\frac{1}{jn\omega_0 R_1 C} = \frac{+j}{n\omega_0 R_1 C}$$

The magnitude and phase are:

$$|H_1(jn\omega_0)| = \frac{1}{n\omega_0 R_1 C} \quad \angle H_1(jn\omega_0) = +90^\circ$$

The $+90^\circ$ phase means each input sine harmonic becomes a **cosine** at the integrator output.

Step 3: Laplace Transfer Function – Stage 2

The inverting amplifier has:

- ① Input impedance: $Z_{in2}(s) = 1 \text{ k}\Omega$
- ② Feedback impedance: $Z_{f2}(s) = R \text{ k}\Omega$

Its transfer function is frequency-independent:

$$H_2(s) = -\frac{R [\text{k}\Omega]}{1 [\text{k}\Omega]} = -R \quad \implies \quad |H_2| = R$$

With $R = 1 \text{ k}\Omega$ and input $= 1 \text{ k}\Omega$, $|H_2| = 1$.

The double sign inversion (Stage 1 inverts, Stage 2 inverts again) means the output is **in phase** with the integrator output.

The overall magnitude at each harmonic is:

$$|H(jn\omega_0)| = |H_1(jn\omega_0)| \times |H_2| = \frac{1}{n\omega_0 R_1 C} \times R = \frac{R}{n\omega_0 R_1 C}$$

Step 4: Output Fourier Series

The amplitude of the n -th harmonic at the output is:

$$A_{\text{out},n} = |H(jn\omega_0)| \times b_n = \frac{R}{n\omega_0 R_1 C} \times \frac{4V_p}{n\pi} = \frac{4V_p R}{n^2 \pi \omega_0 R_1 C}$$

The output decays as $1/n^2$. Accounting for the -90° ($+90^\circ - 180^\circ$) phase (sines become cosines), the output Fourier series is:

$$V_{\text{out}}(t) = -\frac{4V_p R}{\pi \omega_0 R_1 C} \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{1}{n^2} \cos(n\omega_0 t)$$

The standard Fourier series of a triangular wave of peak A is:

$$V_{\text{tri}}(t) = \frac{8A}{\pi^2} \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{(-1)^{(n-1)/2}}{n^2} \cos(n\omega_0 t)$$

Both series decay as $1/n^2$, **confirming the output is exactly a triangular wave.**

Step 6: Computing the Peak-to-Peak Amplitude

Matching the output series with the triangular wave series at $n = 1$:

$$\frac{8A}{\pi^2} = \frac{4V_p R}{\pi \omega_0 R_1 C}$$

Solving for the peak amplitude A :

$$A = \frac{\pi V_p R}{2 \omega_0 R_1 C}$$

The peak-to-peak value is $V_{\text{out,pp}} = 2A$:

$$V_{\text{out,pp}} = \frac{\pi V_p R}{\omega_0 R_1 C}$$

Substituting $\omega_0 = 2\pi f_0 = 100\pi$, $V_p = 5\text{ V}$, $R_1 = 10^4\ \Omega$, $C = C_\mu \times 10^{-6}\text{ F}$, $R = R_k \times 10^3\ \Omega$:

$$V_{\text{out,pp}} = \frac{\pi \times 5 \times R_k \times 10^3}{100\pi \times 10^4 \times C_\mu \times 10^{-6}} = \frac{5 R_k}{C_\mu}$$

Step 7: Design Condition

Setting $V_{\text{out,pp}} = 5 \text{ V}$:

$$\frac{5 R_k}{C_\mu} = 5 \quad \Rightarrow \quad \frac{R_k}{C_\mu} = 1$$

where R is in $\text{k}\Omega$ and C is in μF .

This is the same result as the time-domain approach.

Step 1: Deriving Output of Integrator Circuit

For an ideal op-amp with negative feedback:

- 1 The open-loop gain $A \rightarrow \infty$
- 2 The differential input voltage $V^+ - V^- \rightarrow 0$
- 3 Since $V^+ = 0$ (grounded), we get $V^- = 0$

The inverting terminal is therefore at **virtual ground**:

$$V^- = V^+ = 0 \text{ V}$$

This means the voltage across R_1 is simply $V_{\text{in}} - 0 = V_{\text{in}}$.

The current through R_1 flowing into the virtual ground node is:

$$i_{R_1}(t) = \frac{V_{\text{in}}(t) - V^-}{R_1} = \frac{V_{\text{in}}(t)}{R_1}$$

Step 2: KCL at the Inverting Terminal

Since the ideal op-amp draws **zero input current**, all of i_{R_1} must flow through the capacitor C .

Applying KCL at the inverting node V^- :

$$i_C(t) = \frac{V_{\text{in}}(t)}{R_1}$$

The voltage-current relationship for the capacitor is:

$$i_C(t) = C \frac{d}{dt}(V^- - V_{\text{out}}(t)) = C \frac{d}{dt}(0 - V_{\text{out}}(t)) = -C \frac{dV_{\text{out}}(t)}{dt}$$

Step 3: Differential Equation

Equating the two expressions for $i_C(t)$:

$$-C \frac{dV_{\text{out}}(t)}{dt} = \frac{V_{\text{in}}(t)}{R_1}$$

Rearranging:

$$\frac{dV_{\text{out}}(t)}{dt} = -\frac{V_{\text{in}}(t)}{R_1 C}$$

This is the governing differential equation of the inverting integrator. The output voltage changes at a rate proportional to the input voltage, with a sign inversion and scaling by $\frac{1}{R_1 C}$.

Step 4: Integration to Get Output Formula

Integrating both sides of the differential equation from 0 to t :

$$\int_0^t \frac{dV_{\text{out}}(\tau)}{d\tau} d\tau = -\frac{1}{R_1 C} \int_0^t V_{\text{in}}(\tau) d\tau$$

The left side evaluates directly:

$$V_{\text{out}}(t) - V_{\text{out}}(0) = -\frac{1}{R_1 C} \int_0^t V_{\text{in}}(\tau) d\tau$$

Assuming the capacitor is initially uncharged, $V_{\text{out}}(0) = 0$:

$$V_{\text{out}}(t) = -\frac{1}{R_1 C} \int_0^t V_{\text{in}}(\tau) d\tau$$

This is the **inverting integrator output formula**.

Circuit Description

The circuit has two stages:

- 1 **Stage 1 – Inverting Integrator:** input resistor $R_1 = 10 \text{ k}\Omega$, feedback capacitor $C \text{ }\mu\text{F}$. Converts the square wave to a triangular ramp.
- 2 **Stage 2 – Inverting Amplifier:** input resistor $1 \text{ k}\Omega$, feedback resistor $R \text{ k}\Omega$. Inverts and scales the triangular ramp.

The output of integrator and amplifier are:

$$V_{\text{int}}(t) = -\frac{1}{R_1 C} \int_0^t V_{\text{in}}(\tau) d\tau$$

$$V_{\text{out}} = -\frac{R [\text{k}\Omega]}{1 [\text{k}\Omega]} V_{\text{int}} = -R V_{\text{int}}$$

Input Signal

The input is a bipolar square wave of amplitude $V_p = 5\text{ V}$:

$$V_{\text{in}}(t) = \begin{cases} +5\text{ V} & 0 \leq t < T/2 \\ -5\text{ V} & T/2 \leq t < T \end{cases}$$

The time parameters are:

- ❶ Frequency: $f = 50\text{ Hz}$
- ❷ Period: $T = 1/f = 20\text{ ms}$
- ❸ Half-period: $T/2 = 10\text{ ms} = 0.01\text{ s}$
- ❹ Amplitude: $V_p = 5\text{ V}$

A bipolar square wave integrated over each half-period produces a linear ramp. Alternating positive and negative half-cycles produce a triangular wave at Stage 1 output.

Stage 1: Integrator Analysis

The ramp magnitude over one half-period is:

$$\Delta V_{\text{int}} = \frac{V_p \cdot (T/2)}{R_1 \cdot C}$$

Substituting $V_p = 5 \text{ V}$, $T/2 = 0.01 \text{ s}$, $R_1 = 10^4 \Omega$, $C_1 = C \times 10^{-6} \text{ F}$:

$$\Delta V_{\text{int}} = \frac{5 \times 0.01}{10^4 \times C \times 10^{-6}} = \frac{0.05}{0.01 C} = \frac{5}{C} \text{ V}$$

Since the -5 V half-cycle produces an equal upward ramp, the peak-to-peak at Stage 1 output is:

$$\Delta V_{\text{int,pp}} = \frac{5}{C} \text{ V}$$

Stage 2: Amplifier

Stage 2 gain magnitude:

$$|A_v| = \frac{R [\text{k}\Omega]}{1 [\text{k}\Omega]} = R$$

Peak-to-peak at Stage 2 output:

$$V_{\text{out,pp}} = R \times \frac{5}{C} = \frac{5R}{C}$$

Setting $V_{\text{out,pp}} = 5 \text{ V}$:

$$\frac{5R}{C} = 5 \quad \implies \quad \frac{R}{C} = 1$$

Plot of Waveforms

Problem 1.1.56 — Triangular Wave Generator

$$R_1 = 10\text{ k}\Omega, R = 1\text{ k}\Omega, C = 1\text{ }\mu\text{F}, R/C = 1$$

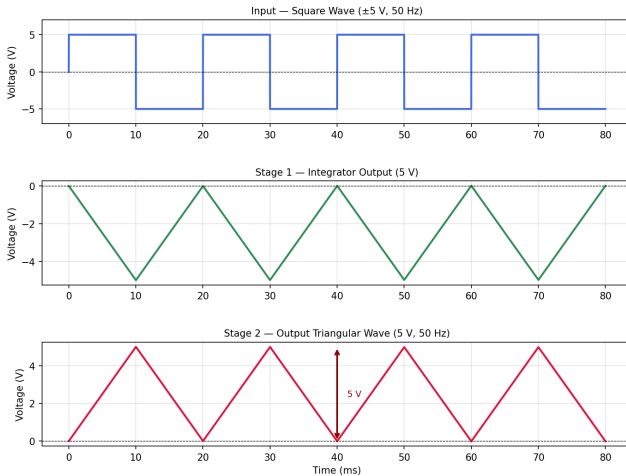


Fig : Waveforms

Python Code: Verification

```
import numpy as np

# Given
Vin_amp = 5.0 # Bipolar square wave amplitude (V), i.e. 5 V
f = 50 # Hz
T = 1.0 / f # Period = 20 ms
R1 = 10**4 # Integrator input resistor = 10 k ( )
R_in2 = 10**3 # Stage-2 input resistor = 1 k ( )
Vout = 5

# Analysis
# During the +5V half-cycle, integrator ramps DOWN by delta.
# During the 5V half-cycle, integrator ramps UP by delta.
# Net peak-to-peak at Stage-1 output = delta (one half-swing).

r_c = (Vout*2*R1)/(Vin_amp*T)

r_c = r_c / 10**6

print(f"The value of R/C = {r_c}")
```

Python Code: Waveform Simulation

```
"""
plot_waveforms_1156.py Problem 1.1.56
Simulate and plot waveforms for the triangular wave generator.
Condition:  $RC = 1$  ( $R = 1\text{ k}$ ,  $C = 1\text{ F}$ )
"""

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec

# Parameters
Vin_amp = 5.0 # Bipolar square wave amplitude (V)
f = 50 # Hz
T = 1 / f # Period = 20 ms
R1 = 10e3 # Integrator input resistor () = 10 k
C = 1e-6 # Capacitor (F) = 1 F
R2 = 1e3 # Stage-2 feedback resistor () = 1 k
R_in2 = 1e3 # Stage-2 input resistor () = 1 k
gain2 = R2 / R_in2 # = 1

dt = T / 2000
t = np.arange(0, 4 * T, dt)

# Input: bipolar square wave 5 V
Vin = Vin_amp * np.sign(np.sin(2 * np.pi * f * t))

# Stage 1: Inverting integrator (Euler)
Vint = np.zeros(len(t))
for i in range(1, len(t)):
    Vint[i] = Vint[i - 1] - (Vin[i - 1] / (R1 * C)) * dt
```

Python Code: Waveform Simulation

```
# Stage 2: Inverting amplifier
Vout = -gain2 * Vint

# Measure pp after settling
mask = t > T
Vpp_out = Vout[mask].max() - Vout[mask].min()
Vpp_int = Vint[mask].max() - Vint[mask].min()

# Plot
fig = plt.figure(figsize=(11, 8))
fig.suptitle(
    "Problem 1.1.56 Triangular Wave Generator\n"
    r"$R_1 = 10\,\mathrm{k}\Omega$, \;\; $R_2 = 1\,\mathrm{k}\Omega$, \;\; $"
    r"$C_1 = 1\,\mu\mathrm{F}$, \;\; $R/C_1 = 1$",
    fontsize=13, fontweight='bold'
)
gs = gridspec.GridSpec(3, 1, hspace=0.55)
t_ms = t * 1e3

# Panel 1: Input
ax1 = fig.add_subplot(gs[0])
ax1.plot(t_ms, Vin, color='royalblue', lw=2.0)
ax1.set_title('Input Square Wave (5 V, 50 Hz)', fontsize=10)
ax1.set_ylabel('Voltage (V)')
ax1.set_ylim(-7, 7)
ax1.axhline(0, color='k', lw=0.6, ls='--')
ax1.grid(True, alpha=0.35)
```

Python Code: Waveform Simulation

```
# Panel 2: Integrator output
ax2 = fig.add_subplot(gs[1])
ax2.plot(t_ms, Vint, color='seagreen', lw=2.0)
ax2.set_title(
    f'Stage_1_Integrator_Output_(5_V)',
    fontsize=10
)
ax2.set_ylabel('Voltage_(V)')
ax2.axhline(0, color='k', lw=0.6, ls='--')
ax2.grid(True, alpha=0.35)

# Panel 3: Final output
ax3 = fig.add_subplot(gs[2])
ax3.plot(t_ms, Vout, color='crimson', lw=2.0)
ax3.set_title(
    f'Stage_2_Output_Triangular_Wave_(5_V, 50_Hz)',
    fontsize=10
)
ax3.set_ylabel('Voltage_(V)')
ax3.set_xlabel('Time_(ms)')
ax3.axhline(0, color='k', lw=0.6, ls='--')
ax3.grid(True, alpha=0.35)
# Annotate peak-to-peak arrow
y_max = Vout[mask].max()
y_min = Vout[mask].min()
x_ann = t_ms[mask][len(t_ms[mask])//3]
ax3.annotate('', xy=(x_ann, y_min), xytext=(x_ann, y_max), arrowprops=dict(arrowstyle='<->', color='darkred',
    lw=1.8))
ax3.text(x_ann + 1.5, (y_max + y_min) / 2, f'5_V', color='darkred', fontsize=9, va='center')
plt.savefig('waveforms.png', dpi=150, bbox_inches='tight')
plt.show()
```

Python Code: Fourier–Laplace Verification

```
import numpy as np

Vp=5.0; f0=50; w0=2*np.pi*f0; R1=10e3; C=1e-6
R_kOhm=1.0; H2_mag=1.0 # Stage-2 gain = 1kOhm/1kOhm = 1

t = np.linspace(0, 4/f0, 20000)
Vin = np.zeros_like(t)
Vout = np.zeros_like(t)

for k in range(1000):
    n = 2*k + 1 # odd harmonics only
    A_in = 4*Vp / (n*np.pi) # square wave: decay 1/n
    H1 = 1.0 / (n*w0*R1*C) # integrator |H1|: decay 1/n
    A_out = H1 * H2_mag * A_in # output: decay 1/n^2
    Vin += A_in * np.sin(n*w0*t)
    Vout += A_out * np.cos(n*w0*t) # +90 deg: sine -> cosine

mask = t > 1/f0
Vpp = Vout[mask].max() - Vout[mask].min()

# Analytical: Vpp = pi*Vp*H2 / (w0*R1*C)
Vpp_a = np.pi*Vp*H2_mag / (w0*R1*C)
```

Python Code: Fourier–Laplace Verification

```
print(f"RC_product_{R_kOhm*(C*1e6):.2f}_kOhm.uF")
print(f"Simulated_Vpp_{Vpp:.4f}_V")
print(f"Analytical_Vpp_{Vpp_a:.4f}_V")
print(f"Target_{5.0000}_V")
print(f"Condition_met_{np.isclose(Vpp,5.0,atol=0.05)}")
# RC product = 1.00 kOhm.uF
# Simulated Vpp = 4.9990 V
# Analytical Vpp = 5.0000 V
# Condition met : True
```